

Project for the Laboratory Part – Instructions

In this project, we will focus on openly-available brain MRI data aiming at devising a general analysis pipeline. The individual data consists in:

- *T1w.nii.gz* → structural T1-weighted image (3D)
- *mask.nii.gz* → brain mask, derived from a brain extraction procedure on the T1w data. This is a binary map, with the same dimension of the corresponding T1w
- *fMRI.nii.gz* → functional MRI data (4D), where each brain volume has been acquired multiple times. This is the fourth dimension of the fMRI data. The TR (repetition time, i.e. time between consecutive time points) is 3.5 sec.

The common anatomical MNI space (MNI-2mm) is also provided (*MNI152_T1_2mm_brain.nii.gz*), along with a reference brain atlas (*Schaefer2018_200Parcels_7Networks_order_FSLMNI152_2mm.nii.gz*) consisting of 200 brain regions derived from a functional parcellation of the brain voxels/tissues (the names of the regions are in the corresponding csv file). They both are 3D and have the same dimensions.

In particular, the pipeline starts with some (basic) processing steps on both T1w and fMRI, and ends with exploratory analyses for feature extraction on the fMRI data. All the steps should be complemented with appropriate visualizations and eventually with summary measures/statistics. If specific filters/methods are used, when possible, try to test the effect of the main parameters varying their values for making appropriate comparisons. The following steps can be followed as general guide:

PART 1 – BASIC PREPROCESSING

This part first concerns data preprocessing of the single-subject *T1w.nii.gz* image. This should include brain extraction (to derive ‘T1_brain’ data, using the provided mask) and tissue segmentation. Moreover, on a single axial slice, perform low-pass filtering (in the frequency domain) and edge detection.

On the 4D fMRI data, apply a gaussian smoothing (on all the volumes), in order to derive ‘fMRI_smoothed’ data to be used in the following steps.

PART 2 – IMAGE REGISTRATION

Using the T1_brain data (3D) and the fMRI_smoothed data (4D) from Part 1, perform the fMRI to MNI linear registration (using FSL commands) following the general scheme fMRI -> T1 -> MNI. The MNI space is provided (*MNI152_T1_2mm_brain.nii.gz*), and this operation needs to be applied on all the fMRI volumes. For each part of the registration, select the most appropriate parameters in terms of cost function, dof and interpolation method.

PART 3 – EXPLORATORY ANALYSES FOR fMRI FEATURE EXTRACTION

Using the spatially normalized fMRI data from Part 2 (i.e., the smoothed 4D fMRI data in MNI space) and the provided brain atlas, compute the mean fMRI time series for each brain parcel. Each time series should have a length corresponding to the total number of time points. The atlas contains 200 regions labeled from 1 to 200, avoiding the background which is coded as 0: this will lead to 200 mean fMRI signals.

For each pair of signals (e.g, mean signals from region A and region B), quantify their similarity, for example relying on Pearson correlation coefficient or on some distance measures. These results can be visualized as similarity matrices, which are often symmetric.

Other analyses could be considered for this exploratory part (e.g, ICA, PCA), directly on the smoothed 4D fMRI data in MNI space, on the extracted time series or on the similarity measures just calculated.

PART 4 – WAVELET ANALYSIS

Select four of the mean fMRI signals extracted from Part 3 and apply a Continuous Wavelet Transform, using a few different continuous wavelets and visualize in all cases the corresponding scalograms. Comment on the findings and the patterns across the four fMRI signals.